

SKILL GAP ANALYSIS FOR JOB APPLICANTS USING DATA ANALYTICS

A PROJECT REPORT

Submitted By

REG NO: 510421205008

G.DEEPIKA

REG NO: 510421205022

S.MAHALAKSHMI

REG NO: 510421205048

S. SASIPRIYA

In partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY

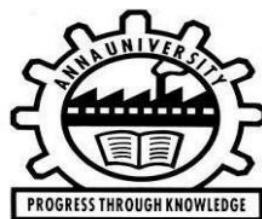
ARUNAI ENGINEERING COLLEGE

(AUTONOMOUS)

TIRUVANNAMALAI– 606 603



ANNA UNIVERSITY: CHENNAI 600 025

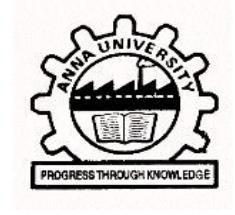


MAY 2025



ANNA UNIVERSITY: CHENNAI 600 025

BONAFIDE CERTIFICATE



Certified that this project report titled **“SKILL GAP ANALYSIS FOR JOB APPLICANTS USING DATA ANALYTICS”** is the bonafide work of **G.DEEPIKA (510421205008), S.MAHALAKSHMI (510421205022) and S.SASIPRIYA (510421205048)**, who carried out the project work under my supervision, for the partial fulfillment of requirements, for the award of degree of Bachelor of Technology in Information Technology. Certified further that to the best of knowledge and belief, work reported herein does not form part of my other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion.

Signature of the Supervisor

Mrs.K.HARIPRIYA, M.E,

Asssistant Prefessor,

Information Technology,

Arunai Engineering College,

Tiruvannamalai-606603.

Signature of the HOD

Dr.K.ANITHA, Ph.D.,

Prof. & HOD,

Information Technology,

Arunai Engineering College,

Tiruvannamalai-606603.



CERTIFICATE OF EVALUATION



COLLEGE NAME : 5104 – ARUNAI ENGINEERING COLLEGE.

BRANCH : B.TECH - INFORMATION TECHNOLOGY

SEMESTER : 8th SEM

SUB CODE & NAME : IT3811-PROJECT WORK

DATE OF EXAM :

NAME OF THE STUDENTS	REG. NO.	TITLE OF THE PROJECT	NAME OF THE INTERNAL GUIDE
G. DEEPIKA	510421205008	Skill Gap Analysis for Job Applicants using Data Analytics	Mrs. K. Haripriya AP/IT
S. MAHALAKSHMI	510421205022		
S.SASIPRIYA	510421205048		

This report is about the **Project Work** submitted for the fulfillment of Bachelor of Technology degree in **INFORMATION TECHNOLOGY** of Anna University and it has been evaluated and confirmed to be report of the work done by the above students. Submitted for the university examination held on _____

Signature of the Supervisor

Mrs. K. HARIPRIYA, M.E,
Assistant professor,
Information Technology,
Arunai Engineering College,
Tiruvannamalai-606603.

Signature of the HOD

Dr.K.ANITHA, Ph.D.,
Prof. & HOD,
Information Technology,
Arunai Engineering College,
Tiruvannamalai-606603.

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

I would like to express my deepest gratitude to all those who have supported and guided me throughout the completion of this final year project.

First and foremost, I am sincerely thankful to **Er. E. V. Kumaran M.E**, Vice Chairman, for his constant encouragement and for providing a nurturing environment at the institution. His leadership and vision have been instrumental in shaping the academic atmosphere in which we thrive.

I extend my heartfelt thanks to **Dr. R. Sathiyaseelan, M.E,Ph.D.**, Registrar, for his unwavering support and for ensuring that all the administrative aspects of our academic journey run smoothly. His dedication to enhancing the learning experience has greatly contributed to the success of this project.

I am deeply grateful to **Dr. C. Elanchezhian, M.E,Ph.D.**, Principal, for his continued support and for fostering an academic environment that encourages innovation and excellence. His guidance has played a crucial role in shaping the educational experiences of all students.

I would also like to express my sincere thanks to **Dr. K. Anitha, M.E,Ph.D.**, Head of the Department, for her invaluable support and for providing the resources and guidance necessary to complete this project. Her leadership has greatly enriched my academic experience.

A special note of gratitude goes to **Mrs. K. Sahitya Priyadharshini, M.E,(Ph.D)**, Project Coordinator, for her continuous encouragement, guidance, and for ensuring the smooth execution of the project. Her expertise and attention to detail have been incredibly helpful.

I am extremely fortunate to have had **Mrs. K. Haripriya, M.E**, as my project supervisor. Her mentorship, patience, and insightful suggestions were pivotal in refining my ideas and pushing me to achieve the best possible outcome for this project. Her dedication to helping students succeed is deeply appreciated.

Finally, I would like to express my gratitude to all my friends, classmates, and family members for their unwavering support and encouragement throughout this journey.

Thank you all for being a part of this significant achievement.

ABSTRACT

The project titled Skill Gap Analysis for Job Applicants is a data analytics-driven solution designed to bridge the gap between the skills possessed by job seekers and the skills required for various job roles. This system utilizes Natural Language Processing (NLP) techniques to process and interpret unstructured skill data submitted by users. To analyze and categorize job roles, K-Means Clustering is applied to group similar roles based on required skill sets, while the K-Nearest Neighbors (KNN) algorithm is employed to recommend the most relevant job role corresponding to the user's existing skills. An interactive web interface is developed using Streamlit, facilitating user-friendly input and real-time feedback. Data visualizations generated using Matplotlib provide insights into the clustering of job roles and illustrate the alignment between user skills and job requirements. A key feature of this system is its ability to identify missing skills for each recommended job role. For every identified gap, the system offers personalized learning resources sourced from platforms such as Coursera, enabling users to upskill effectively and enhance their employability.

திட்ட சுருக்கம்

வேலைத் திறன் சிக்கல் பகுப்பாய்வு என்பது, பயனாளரின் தற்போதைய திறன்கள் மற்றும் வேலைக்கு தேவையான திறன்கள் இடையிலான வேறுபாட்டை கண்டறிந்து, அவர்களுக்கு பொருத்தமான வேலை வாய்ப்புகளை பரிந்துரைக்கும் ஒரு தரவுப் பகுப்பாய்வு தீர்வு ஆகும். இதில் NLP தொழில்நுட்பம் மூலம் பயனாளரின் திறன்கள் செயலாக்கப்படுகின்றன, K-Means கிளஸ்டரிங் மூலம் வேலைத் திறன்கள் வகைப்படுத்தப்படுகின்றன, மேலும் KNN அல்கொரிதம் மூலம் பொருந்தக்கூடிய வேலைகள் பரிந்துரைக்கப்படுகின்றன. Streamlit பயன்படுத்தி எளிய இணையதளம் உருவாக்கப்பட்டுள்ளது மற்றும் Matplotlib மூலம் தரவுகள் காட்சியளிக்கப்படுகின்றன. இல்லாத திறன்களை மேம்படுத்த, Coursera போன்ற தளங்களிலிருந்து கற்றல் பரிந்துரைகள் வழங்கப்படுகின்றன. இந்த அமைப்பு மாணவர்கள், புதிய பட்டதாரிகள் மற்றும் தொழில்முனைவோர்களுக்கு மிகவும் பயனுள்ளதாகும்.

TABLE OF CONTENTS

Chapter No.	Title	Page No.
	Bonafide Certificate	i
	Certificate of Evaluation	ii
	Acknowledgement	iii
	Abstract(English)	iv
	Abstract(Tamil)	v
	List of Figures	viii
	List of Abbreviations	ix
1	INTRODUCTION	1
	1.1 Purpose of the System	2
	1.2 Objective of the Project	3
	1.3 Scope of the Project	4
2	SYSTEM ANALYSIS	5
	2.1 Existing System	5
	2.2 Proposed System	7
3	LITERATURE SURVEY	9
	3.1 Skill Match Analyzer	9
	3.2 CareProfSys: A Job Recommender System Based on ML and Ontology	10
	3.3 Enhancing Job Recommendations Using NLP and ML Techniques	11
4	SYSTEM REQUIREMENTS	13
	4.1 System Requirement Specification	13
	4.2 Hardware Requirements	14
	4.3 Software Requirements	14
	4.4 Software Descriptions	15

5	SYSTEM DESIGN	17
	5.1 System Architecture	17
	5.2 ER Diagram	18
	5.3 Data Flow Diagram	19
	5.4 Use Case Diagram	19
	5.5 Cluster Graph (For Visualization)	20
6	SYSTEM IMPLEMENTATION	21
	6.1 Natural Language Processing(NLP)	21
	6.2 K-Means Clustering	22
	6.3 K-Nearest Neighbours(KNN)	23
	6.4 Streamlit for Web Development	23
	6.5 Matplotlib for Visualization	24
7	METHODOLOGY	25
	7.1 Data Collection and Preprocessing	25
	7.2 Model Implementation(K-Means and KNN)	28
	7.3 Upskilling Resources	30
	7.4 Website Implementations	31
8	SAMPLE CODING & SCREENSHOTS	32
	8.1 Sample Coding	32
	8.2 Screenshots	43
9	CONCLUSION AND FUTURE ENHANCEMENT	47
	9.1 Conclusion	47
	9.2 Future Enhancement	48
10	REFERENCES	49
	10.1 References	49
11	CHART	51
	11.1 Gantt Chart	51

LIST OF FIGURES

S.NO	NAME OF THE FIGURES	PAGE NO
1	5.1.1 System Architecture	17
2	5.2.1 ER Diagram	18
3	5.3.1 Data Flow Diagram	19
4	5.4.1 Use Case Diagram	19
5	5.5.1 Cluster Graph (For Visualization)	20

LIST OF ABBREVIATIONS

S.NO	ABBREVIATIONS	FULLFORM
1	ML	Machine Learning
2	KNN	K Nearest Neighbours
3	BERT	Bidirectional Encoder Representations from Transformers.
4	UI	User Interface
5	TF-IDF	Tern Frequency-Inverse Document Frequency
6	API	Application Programming Interface
7	NLP	Natural Language Processing
8	ER	Entity Relationship
9	SVD	Singular Value Decomposition
10	IDE	Integrated Development Environment